

Assignment 3

- Prove that in a heap, with nodes indexed by $1, \dots, n$, starting with the root, the node with largest index to have children, has index $\lfloor n/2 \rfloor$.
- Is the operation of deletion in a binary search tree “commutative”?

That is, would deleting x and then y , yield the same tree as deleting y and then x ?

Argue why it is commutative, or give a counterexample.

- Suppose that you construct a binary search tree by repeatedly inserting distinct values into the tree.

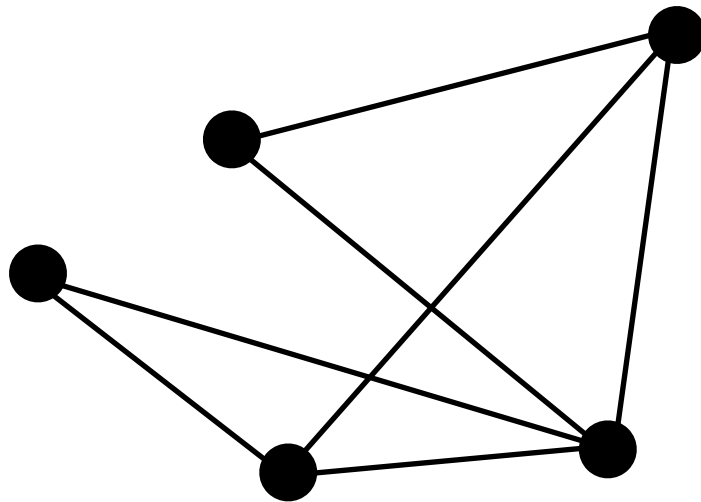
Argue that the number of nodes examined in searching for a value in the tree is 1 plus the number of nodes examined when the value was first inserted into the tree.

- The square of a directed graph $G = (V, E)$ is the graph $G^2 = (V, E^2)$ such that $(u, v) \in E^2$ if and only if G contains a walk with two edges between u and v .

Describe an efficient algorithm for computing G^2 from the adjacency matrix of G , and analyze the running time.

- Most graph algorithms that take an adjacency-matrix representation as input require $\Omega(V^2)$ time, but there are some exceptions. Show how to determine whether a directed graph contains a universal sink - a vertex with in-degree $|V| - 1$ and out-degree 0 in $O(V)$ time, given the adjacency matrix.

- Using matrix multiplication, compute the number of walks on 3 edges between all pairs of vertices, for the graph below. Your answer should be a square matrix on the number of vertices.



- Give an example of a directed graph, $G=(V, E)$, a source vertex s , and a subset of edges which form a tree, $E_\pi \subseteq E$, with $V(G|_{E_\pi}) = V$. Such that for each vertex, v , the unique path in the graph (V, E_π) from s to v , is a shortest path in G , yet the set of edges E_π cannot be produced by running BFS on G , no matter how the vertices are ordered in the adjacency lists.
- The diameter of a tree is defined as the largest of all distances in the tree. Give an efficient algorithm to compute the diameter of a tree, and analyze the running time of your algorithm.